

Finite symmetry at measurable strength

Prikry deletion, definable selectors,
and countable ultrafilter families

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Abstract. The supplied synthesis asks whether its finite-selector obstruction and the necessity direction of its finite-label classification already hold in a Prikry extension of a measurable cardinal. We prove that they do, even for objects definable from the finite-change class of the generic sequence together with arbitrary ground-model set parameters. The proof replaces an elementary embedding by a one-point deletion after a fixed Prikry stem. Pure decision makes a finite observation constant, while deletion rotates the observation. Encoding a whole finite family of selectors by its finite restriction tables permits the cycle-amplification argument of the synthesis to be transferred to forcing.

We also determine a sharp local threshold for ultrafilters. On the finite-change class of the Prikry sequence there is no nonempty finite family of ultrafilters definable from the permitted parameters, but there is a definable countably infinite family. More generally, every cofinal finite-change class at any uncountable cardinal of countable cofinality carries a countable, definable-from-a-low-rank-parameter family of ultrafilters, naturally organized as a free transitive integer orbit. Finally, the combined finite-symmetry, normal-trace and sharp local-width package is equiconsistent with one measurable cardinal. The matching lower bound comes from its explicit normal-measure inner-model clause, not from the selector obstruction alone.

The new result relative to the supplied reports

The finite-symmetry phenomena previously obtained from ultraexactness also occur after ordinary Prikry forcing. They therefore do not, by their joint consistency alone, distinguish the ultraexacting/I0 level from the measurable level. The new step is the *finite-observation deletion argument*, not a new proof of the Prikry property or a new finite-permutation lemma.

The local ultrafilter threshold is exactly $\aleph_0$. A countable-valued assignment of ultrafilters is not a countable family of global ultrafilter-valued kernels; that latter question remains separate.

Scope and status. This is a proof-based continuation of Cardinals4.zip, principally [1, §§9, 12–13, 15]. The main forcing arguments below are self-contained, apart from the general forcing theorem and standard facts about relativized HOD, whose use is explained. No new inconsistency of exacting or ultraexacting cardinals is claimed. The proposed theorems have been checked by re-deriving the arguments and by finite computational tests of the combinatorial parts; they have not been independently refereed or formalized. No claim of literature priority is made.

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1 Main results and their significance

For an uncountable cardinal λ of cofinality ω , let

$$D_\lambda = \{a \subseteq \lambda : \text{ot}(a) = \omega, \sup a = \lambda\}, \quad Q_\lambda = D_\lambda / \equiv^*,$$

where $a \equiv^* b$ means that $a \Delta b$ is finite. The quotient is always computed in the ambient universe. Write

$$\text{Sel}_{n,r}(Q) = \{f : [Q]^n \longrightarrow [Q]^r : f(B) \subseteq B \text{ for every } B \in [Q]^n\} \quad (1 \leq r < n < \omega).$$

A member is an r -out-of- n selector. Choice guarantees that these sets are nonempty. Our conclusions concern *definability*, not their nonemptiness.

Theorem 1.1 (Prikry finite symmetry). Let $W \models \text{ZFC}$, let $U \in W$ be a normal measure on a measurable cardinal κ , and let $V = W[c]$ be the extension by $\mathbb{P}_U$, with c the increasing generic sequence. Put $q_0 = [\text{ran}(c)] \in Q_\kappa^V$. Then:

1. For all $1 \leq r < n < \omega$, there is no nonempty finite

$$\mathcal{F} \subseteq \text{Sel}_{n,r}(Q_\kappa)$$

that is ordinal definable in V from q_0 and finitely many set parameters belonging to W .

2. Let E be a finite set, coded by a natural number, with an action of S_m . An equivariant rule

$$L : \text{Inj}(m, Q_\kappa) \longrightarrow E$$

definable from such parameters exists if and only if every $\pi \in S_m$ fixes at least one element of E . In the positive direction a parameter of rank below $\omega + 10$ suffices.

3. There is no nonempty finite family of ultrafilters on q_0 definable from the same parameters. Nevertheless there is a countably infinite such family, definable from q_0 and a nonprincipal ultrafilter on ω belonging to W .

Here $\text{Inj}(m, Q)$ is the set of injective ordered m -tuples. The action on tuples is

$$(\pi \cdot \vec{q})_i = q_{\pi^{-1}(i)}, \quad L(\pi \cdot \vec{q}) = \pi \cdot L(\vec{q}).$$

Using representatives instead of classes yields the same statement provided the rule is invariant under changing finitely many points in each coordinate.

Corollary 1.2 (The low-rank version). Every conclusion of Theorem 1.1 holds with “definable from q_0 and ground-model parameters” replaced by $\text{OD}_{V_\kappa \cup \{q_0\}}$. In particular, the nonexistence assertions apply to OD_{V_κ} families and rules. The finite-label equivalence also holds with low-rank parameters alone; the positive local ultrafilter construction still uses q_0 .

Proof. Prikry forcing adds no sets of rank below κ ; a proof appears in Proposition 3.5. Hence every permitted low-rank parameter belongs to W . $\square$

The synthesis already proves the finite algebra used for (1), and the positive direction of (2), in its ultraexacting analysis. It specifically leaves the Prikry-extension analogue open. We reuse and prove that algebra, but supply a different mechanism that realizes it. This establishes the *Prikry-extension assertion* in the last question of [1, §15], item (ii). It does *not* show that the conclusions follow from the abstract rigidity of a class alone.

Theorem 1.3 (Sharp countable ultrafilter families, without forcing). Let λ be any uncountable cardinal of cofinality ω , and fix a nonprincipal ultrafilter u on ω . There is a function, definable from u and λ , assigning to every $q \in Q_\lambda$ a countably infinite family $\mathcal{U}_u(q)$ of nonprincipal ultrafilters on q . This family admits a canonical free transitive $\mathbb{Z}$ -action.

This gives a useful new sharpness statement even in the original ultraexacting setting: at each class for which the synthesis excludes finite definable families of ultrafilters, the least possible size of a nonempty $\text{OD}_{V_\lambda \cup \{q\}}$ family is exactly $\aleph_0$.

Theorem 1.4 (Consistency calibration). The finite-symmetry and normal-trace package FSNT defined in Section 10 is equiconsistent over ZFC with one measurable cardinal. It includes the selector and finite-label conclusions, rigidity of a cofinal class, its sharp countable ultrafilter width, and a normal tail measure on λ in an inner model containing all of V_λ .

This calibrates a specific package, not ultraexactingness itself. The substantially stronger consistency analysis of ultraexacting cardinals in [4, 5] is not changed.

2 Parameters, quotient classes, and forcing conventions

2.1 Definability does not require hereditary membership

For a set of permitted parameters A , OD_A consists of sets definable from finitely many members of A and finitely many ordinal parameters. When we write $\text{OD}_{p,q}$, the objects p and q are each available as a *single parameter*; their elements are not automatically available. The inner model $\text{HOD}_{p,q}$ consists of the sets all of whose transitive-closure members, as well as the sets themselves, are in $\text{OD}_{p,q}$.

In $V = W[c]$, the notation $\text{OD}_{W \cup \{q_0\}}$ is external shorthand: each definition uses q_0 , a fixed finite tuple of ground-model sets, and ordinal parameters. It does not mean that a truth predicate for W is added to the language. A tuple of ground-model sets may be coded into one ground-model set p .

A finite family can be definable without any particular member being definable. Consequently the proof must not select a canonical member of such a family. Restriction tables will avoid that mistake.

2.2 Normal measures and conditions

Work first in W . A normal measure U on κ is a nonprincipal κ -complete ultrafilter on κ satisfying normality: every regressive function on a member of U is constant on a member of U . Normality implies that diagonal intersections of κ many members of U are in U , and that every club in κ is in U .

A condition in $\mathbb{P}_U$ is (s, A) , where s is a finite increasing sequence from κ , $A \in U$, and every member of A is above the last point of s . The stronger-condition convention is

$$(t, B) \leq (s, A)$$

if t extends s , every point of t added after s lies in A , and $B \subseteq A$. A *direct* or *pure* extension, written $\leq^*$, leaves the stem unchanged. Intersections give lower bounds for direct descending sequences of length less than κ .

For a generic filter G , the union of the stems is an increasing ω -sequence c_G . Conversely the sequence determines the filter. For a condition (s, A) we write $c \Vdash (s, A)$ when s is an initial segment of c and all the subsequent points of c lie in A .

The following proofs use the ordinary forcing theorem: forcing is definable in the ground model, the truth lemma holds, and isomorphic cones transport generic filters. The expository language of transitive ground models is convenient; the consistency implications are the usual syntactic forcing implications and do not assume that mere consistency supplies a transitive model.

3 A proved Prikry toolkit

The results of this section are classical; see [3, §3]. We include the proofs of precisely the ingredients needed later. No iterability theorem or elementary embedding is used in the new forcing argument.

3.1 Diagonalization and finite homogeneity

Lemma 3.1 (Diagonal intersections). If $\langle A_\xi : \xi < \kappa \rangle$ is a sequence of members of U , then

$$\Delta_{\xi < \kappa} A_\xi = \{\beta < \kappa : \forall \xi < \beta (\beta \in A_\xi)\} \in U.$$

Consequently, given a member $B_t \in U$ for every finite increasing sequence t from κ , there is a set $B \in U$ of limit ordinals such that

$$t \subseteq B, \quad \beta \in B, \quad \max t < \beta \implies \beta \in B_t. \quad (*)$$

The empty sequence can be included in this requirement.

Proof. If the diagonal intersection is not in U , on its complement choose the least $\xi < \beta$ witnessing $\beta \notin A_\xi$. This is a regressive function. Normality makes it constantly ξ on a member of U , contradicting $A_\xi \in U$.

Every club $C \subseteq \kappa$ belongs to U . Otherwise, after deleting a bounded initial segment, the function $\beta \mapsto \sup(C \cap \beta)$ is regressive on $\kappa \setminus C$; closedness of C gives strict inequality. Normality would make it constant on a measure-one set, whereas unboundedness of C makes every constant fiber bounded. Bounded sets cannot belong to a nonprincipal κ -complete ultrafilter. This is a contradiction.

For the second assertion, put

$$C_\alpha = \bigcap \{B_t : t \text{ is a finite increasing sequence from } \alpha\}.$$

There are fewer than κ sequences in this intersection, so $C_\alpha \in U$. Intersect $\Delta_\alpha C_\alpha$ with the club of nonzero limit ordinals and with $B_\emptyset$. If $\max t < \beta$ and β is a limit ordinal, choose α with $\max t < \alpha < \beta$. Then $t \subseteq \alpha$ and $\beta \in C_\alpha \subseteq B_t$. $\square$

Lemma 3.2 (Finite-dimensional homogeneity). If $\rho < \kappa$ and $h : [\kappa]^n \rightarrow \rho$, then some $A \in U$ is homogeneous for h . Given one such coloring for every $n < \omega$, one can choose a single member of U homogeneous separately at every length.

Proof. The case $n = 0$ is immediate. For $n = 1$, a partition into fewer than κ cells has a unique cell in U . Suppose the assertion holds for n . For each $t \in [\kappa]^n$, choose a color $\varepsilon(t) < \rho$ and $B_t \in U$ such that

$$h(t \frown \langle \beta \rangle) = \varepsilon(t) \quad \text{for } \beta \in B_t \text{ above } \max t.$$

This is possible by κ -completeness and the ultrafilter property. Apply Lemma 3.1, then the induction hypothesis to the coloring $t \mapsto \varepsilon(t)$. On the intersection of the two resulting measure-one sets, h is constant on $(n + 1)$ -tuples. Finally intersect the countably many homogeneous sets to treat all finite lengths simultaneously. $\square$

3.2 Pure decision

Theorem 3.3 (Prikrý property). For every forcing sentence φ and every $p = (s, A)$, there is a direct extension of p deciding φ . If a name is forced below p to belong to a fixed finite ground-model set, a direct extension decides its value.

Proof. For each finite extension t of the stem using points of A , ask whether some reservoir $B_t \in U$ makes $(s \cap t, B_t)$ decide φ . If so, choose one and color t by the corresponding truth value 0 or 1. Two opposite decisions with the same stem are impossible, since their reservoirs have measure-one intersection. If no such reservoir exists, give t the third color 2 and choose an arbitrary measure-one reservoir.

Diagonalize the chosen reservoirs using Lemma 3.1, retaining a set $B \subseteq A$ in U . Thus whenever a finite $t \subseteq B$ has color 0 or 1, the condition

$$(s \cap t, B \setminus (\max t + 1))$$

forces that value. The empty-stem case is covered by including its reservoir in the intersection. By Lemma 3.2, shrink to $C \in U$ so that the color is constant separately on each $[C]^n$.

Below (s, C) some condition decides φ , by the forcing theorem. Its finite added stem has some length n and color $\varepsilon \in \{0, 1\}$. Hence every n -point extension using C has color ε and forces the same value with reservoir C above its stem.

These n -point conditions are predense below (s, C) . Indeed, any longer stem extends one of them; any shorter stem can be lengthened to n points from its reservoir. Consequently conditions forcing value ε are dense below (s, C) , so (s, C) itself decides φ .

For a name valued in a fixed finite set, apply the assertion successively to its possible values, retaining the stem. The final direct extension decides every equality; at least one must hold. $\square$

Remark 3.4. We never infer that an arbitrarily chosen direct extension belongs to the originally given generic filter. In contradiction arguments we first work below a condition that forces the proposed object to exist, choose a deciding direct extension, and then use a generic through that stronger condition. This distinction is essential.

3.3 Preservation and stem surgery

Proposition 3.5 (Preservation). Prikrý forcing preserves all cardinals, adds no bounded subsets of κ , and satisfies

$$V_\kappa^{W[c]} = V_\kappa^W, \quad W[c] \models \text{cf}(\kappa) = \omega \text{ and } \kappa \text{ is a strong limit cardinal.}$$

Proof. If a name is forced to be a subset of $\alpha < \kappa$, decide its membership statements by a direct descending recursion of length α , using Theorem 3.3. Intersections at limit stages and at the end exist by κ -completeness. The final direct extension decides the entire subset. Thus no bounded subsets of κ are added.

Cardinals below κ are preserved: a collapse between ordinals below κ would be coded by a new bounded subset of κ , using a ground-model pairing function. Since κ is a limit of ground-model cardinals, it remains a cardinal. Conditions with the same stem are compatible, and there are only κ stems. Hence $\mathbb{P}_U$ is κ^+ -cc and preserves all cardinals at least κ^+ . For completeness, the relevant chain-condition argument bounds the possible values of each coordinate of a name by an antichain of

size at most κ ; a purported collapse of a larger cardinal would therefore already supply an impossibly small ground-model cover of it.

There are no new subsets of any ground-model set of size below κ , by coding with an ordinal below κ . Since κ was inaccessible, induction on $\alpha < \kappa$ now gives $V_\alpha^{W[c]} = V_\alpha^W$. The successor step uses $|V_\alpha^W| < \kappa$, and the limit step is a union. This proves rank agreement. The same preservation of small power sets proves that κ remains strong limit.

The generic sequence has order type ω , since stems are finite and their lengths are unbounded. For every $\beta < \kappa$, extending the stem above β is dense. Thus its supremum is κ . $\square$

Lemma 3.6 (Isomorphic cones and deletion). Let s, t be finite increasing stems and let $A \in U$ lie above every point of both stems. The map

$$(s \frown u, B) \longmapsto (t \frown u, B) \quad (\dagger)$$

is an isomorphism between the cones below (s, A) and (t, A) . Corresponding generic sequences differ at finitely many points, give the same forcing extension, and have the same class in Q_κ .

In particular, if $d \Vdash (s, A)$ is generic and $k = |s|$, then the sequence obtained by deleting d_k is also generic, satisfies (s, A) , and generates $W[d]$.

Proof. The common finite suffix u consists of points from A , above both stems. The displayed map and its inverse preserve and reflect the order. They belong to W , so they transport generic filters, and the filters recover each other. Their sequences have the same infinite tail. All finitely many differing ordinals belong to W , hence either sequence recovers the other over W .

For the last assertion put $\beta = d_k$. Work below

$$(s \frown \langle \beta \rangle, A \setminus (\beta + 1)).$$

The original generic passes through this condition. Replace its stem by s using the cone isomorphism. The resulting generic is exactly d with β deleted. Its remaining tail is still contained in A , so it satisfies the original (s, A) . $\square$

This proof is sufficient for every finite modification used below. We do not need to invoke the full Mathias criterion for arbitrary sequences.

4 The finite-observation deletion principle

Lemma 4.1 (Deletion forces a finite fixed point). Suppose a condition p forces the existence of an object z uniquely defined in the extension from a ground-model parameter a , ordinal parameters, and the generic class $[\text{ran}(\dot{c})]$. Let $E \in W$ be finite, and suppose a specified construction gives

$$T(d, z) \in E$$

for every generic d through p . Let ρ be a permutation of E . Assume that deleting the first point after a fixed stem, while retaining that stem, always changes this observation according to

$$T(d^-, z) = \rho(T(d, z)). \quad (1)$$

Then ρ has a fixed point in E .

Proof. The construction determines a finite-valued forcing name. By pure decision choose $p^* = (s, A) \leq^* p$ and $e \in E$ such that p^* forces this name to equal e . Take a generic sequence d through p^* and delete its first point after s , obtaining d^- . By Lemma 3.6, both sequences are generic through p^* and $W[d] = W[d^-]$. Their finite-change classes coincide, and all the remaining parameters are ground-model sets or ordinals. Therefore the unique object called z is *the same object* in the two evaluations. Both observations equal e , while (1) makes the second $\rho(e)$. Hence $e = \rho(e)$. $\square$

Remark 4.2 (Why definability is used). We do not apply an automorphism to the completed universe. The universe has not changed at all: only its generic presentation has changed. Definability ensures that the name of the proposed selector, family, or rule evaluates to the same object in these two presentations. A parameter equal to the distinguished representative d would invalidate this step; a parameter equal to its class $[\text{ran}(d)]$ does not.

One can use the lemma when E is a finite power set or a finite space of tables. It is not necessary that the observation be a single selected point. This is the key to handling a finite definable family with no definable member.

5 Finite families of selectors

5.1 The finite algebra

Let $A_N = \{0, \dots, N-1\}$ and let $\tau(i) = i+1 \pmod{N}$. Let

$$\mathcal{T}_{N,n,r} = \{v : [A_N]^n \rightarrow [A_N]^r : v(B) \subseteq B\}.$$

The cyclic action is

$$(\tau \cdot v)(B) = \tau''v(\tau^{-1}''B).$$

This is a finite set with a permutation action.

Lemma 5.1 (Cycle amplification; from the synthesis). Fix $1 \leq r < n$, $M \geq 1$, put

$$L = \text{lcm}(1, 2, \dots, M), \quad N = nL.$$

There is no nonempty τ -invariant $\mathcal{A} \subseteq \mathcal{T}_{N,n,r}$ with $|\mathcal{A}| \leq M$.

Proof. If such $\mathcal{A}$ existed, τ would permute it. Each orbit length would be at most M , hence divide L . Thus τ^L would fix every $v \in \mathcal{A}$.

The set

$$B_0 = \{0, L, 2L, \dots, (n-1)L\}$$

is invariant under τ^L , and τ^L acts on it as one n -cycle. For any $v \in \mathcal{A}$ we would have

$$v(B_0) = v(\tau^L''B_0) = \tau^L''v(B_0).$$

A subset invariant under a transitive cycle is either empty or the entire cycle. But $|v(B_0)| = r$ and $0 < r < n$, a contradiction. $\square$

For $M = 2$, $n = 2$, $r = 1$, one must use a four-cycle. A two-cycle alone does not exclude a pair of selectors that it interchanges. This is why merely averaging, or arbitrarily choosing one member of a finite family, would leave a genuine gap.

5.2 Tables obtained from a generic sequence

For a generic increasing sequence d , define

$$a_i(d) = \{d_{Nk+i} : k < \omega\}, \quad q_i(d) = [a_i(d)] \quad (i < N). \quad (2)$$

The $q_i(d)$ are distinct classes, with pairwise disjoint cofinal representatives. If d^- is obtained by deleting one point, then after ignoring finitely many terms

$$q_i(d^-) = q_{i+1 \bmod N}(d). \quad (3)$$

This remains true when the point is deleted after an arbitrarily long fixed stem: the finite prefix is irrelevant in Q_κ .

Suppose $\mathcal{F}$ is a finite family of r -out-of- n selectors. For $f \in \mathcal{F}$, define a finite table

$$v_{f,d}(B) = \{i \in B : q_i(d) \in f(\{q_j(d) : j \in B\})\} \quad (B \in [A_N]^n).$$

Set

$$\mathcal{A}(d, \mathcal{F}) = \{v_{f,d} : f \in \mathcal{F}\} \subseteq \mathcal{T}_{N,n,r}. \quad (4)$$

Different selectors may have identical restrictions, so only $1 \leq |\mathcal{A}(d, \mathcal{F})| \leq |\mathcal{F}|$ is asserted.

Theorem 5.2 (No finite definable selector family). Under the hypotheses of Theorem 1.1, no nonempty finite $\mathcal{F} \subseteq \text{Sel}_{n,r}(Q_\kappa)$ belongs to $\text{OD}_{W \cup \{q_0\}}$.

Proof. Suppose otherwise. Fix a formula and permitted parameters defining $\mathcal{F}$ uniquely. By the truth lemma choose a condition p forcing that this formula defines a nonempty finite family of the required kind; strengthen if necessary so that its size is the fixed integer $M \geq 1$. Define L and N as in Lemma 5.1.

The name for the set of tables in (4) takes values in the finite ground-model set $\mathcal{P}(\mathcal{T}_{N,n,r})$. By pure decision choose $p^* = (s, A) \leq^* p$ forcing

$$\mathcal{A}(\dot{c}, \dot{\mathcal{F}}) = \mathcal{A}_*.$$

Here $\mathcal{A}_*$ is nonempty and has at most M elements.

Take a generic d through p^* and delete the first point after s . The family $\mathcal{F}$ is the same in $W[d]$ and $W[d^-]$, by Remark 4.2. Equation (3) gives, for each f and B ,

$$v_{f,d^-}(B) = \tau^{-1} v_{f,d}(\tau B).$$

Consequently

$$\mathcal{A}(d^-, \mathcal{F}) = \tau^{-1} \cdot \mathcal{A}(d, \mathcal{F}).$$

Both observations equal the decided value $\mathcal{A}_*$. Hence $\mathcal{A}_*$ is τ -invariant. This contradicts Lemma 5.1. $\square$

Corollary 5.3 (Orders and choice functions). There is no nonempty finite $\text{OD}_{W \cup \{q_0\}}$ family of linear orders on Q_κ , tournaments on Q_κ , or choice functions on its nonempty finite subsets. In particular Q_κ admits no injection into the ordinals definable from those parameters.

Proof. A linear order gives a binary selector by taking the smaller point; a tournament gives one by taking the winner; a finite-set choice function restricts to one. Applying the appropriate definable conversion to a nonempty finite family yields a nonempty finite family of binary selectors. An injection into the ordinals gives a definable linear order. Theorem 5.2 excludes all of them. $\square$

The conclusion concerns the complete ambient Q_κ , not just the finitely many probe classes. The probe classes serve only to extract a finite contradiction from a purported global object.

6 The exact finite-label classification

6.1 Realizing an arbitrary permutation by deletion

Lemma 6.1 (Permutation probes). Fix $m < \omega$ and $\pi \in S_m$. From a Prikry generic d one can define an injective tuple $\vec{q}(d) \in \text{Inj}(m, Q_\kappa)$ such that deleting one point of d gives

$$q_i(d^-) = q_{\pi(i)}(d) \quad (i < m). \quad (5)$$

Proof. After strengthening the reservoir, all sufficiently late points of d are limit ordinals. Define

$$a_i(d) = \{d_n + \pi^{-n}(i) : n < \omega\}, \quad q_i(d) = [a_i(d)].$$

For sufficiently large n , the finite blocks $\{d_n + i : i < m\}$ lie strictly below d_{n+1} and are mutually disjoint. Within a block the values $\pi^{-n}(i)$ are distinct. Thus the a_i are cofinal ω -sets and their classes are pairwise distinct. Any exceptional initial intersections are finite and immaterial.

After deleting the point in position k , the new sequence satisfies $d_n^- = d_{n+1}$ for $n \geq k$. For these n ,

$$d_n^- + \pi^{-n}(i) = d_{n+1} + \pi^{-(n+1)}(\pi(i)).$$

The two corresponding representatives differ at only finitely many points, proving (5). $\square$

Theorem 6.2 (Necessity in a Prikry extension). Suppose $L : \text{Inj}(m, Q_\kappa) \rightarrow E$ is equivariant and belongs to $\text{OD}_{W \cup \{q_0\}}$, where E is a finite coded S_m -set. Then

$$\text{Fix}_E(\pi) \neq \emptyset \quad \text{for every } \pi \in S_m. \quad (6)$$

Proof. Fix π . As in Theorem 5.2, work below a condition forcing that the chosen formula defines the proposed rule. Use the tuple from Lemma 6.1 and make the finite-valued observation

$$T(d) = L(\vec{q}(d)).$$

Pure decision fixes its value below a direct extension (s, A) . Deletion after s replaces $\vec{q}(d)$ by $\pi^{-1} \cdot \vec{q}(d)$, by (5). The rule L itself does not change. Equivariance therefore gives

$$T(d^-) = \pi^{-1} \cdot T(d).$$

Lemma 4.1 yields a fixed point of π^{-1} , equivalently of π . $\square$

6.2 Why the condition is sufficient

The following direction is combinatorial and does not need forcing or a large cardinal. It is the cyclic-stabilizer argument of [1, §12.5], included to make the classification complete.

Lemma 6.3 (Cyclic stabilizers of incidence words). Let $w = (w_n)_{n < \omega}$ be a word over $\mathcal{A}_m = \mathcal{P}(m) \setminus \{\emptyset\}$. Assume that its columns are pairwise eventually different: for $i \neq j$, infinitely many n satisfy

$$(i \in w_n) \leftrightarrow (j \in w_n).$$

Let $[w]_{\text{tail}}$ identify words whose tails agree after possibly different finite initial segments. Then

$$H_w = \{\pi \in S_m : [\pi w]_{\text{tail}} = [w]_{\text{tail}}\}$$

is cyclic.

Proof. Let P_w be the subgroup of $\mathbb{Z}$ consisting of all eventual periods:

$$P_w = \{k \in \mathbb{Z} : w_{n+k} = w_n \text{ for all sufficiently large } n\}.$$

Negative indices cause no problem because only sufficiently large n are used. Either $P_w = \{0\}$ or $P_w = d\mathbb{Z}$ for a positive integer d .

For $\pi \in H_w$, there is an integer k such that $\pi w_n = w_{n+k}$ eventually. The residue class of k modulo P_w is unique. Assigning this residue to π is a homomorphism

$$H_w \longrightarrow \mathbb{Z}/P_w :$$

composition of permutations adds the offsets, because permutations of letters commute with shifts of positions.

The homomorphism is injective. If the offset is zero modulo P_w , then $\pi w_n = w_n$ eventually. For every i , membership of i and of $\pi(i)$ in all sufficiently late letters is therefore identical. Distinctness of the columns implies $\pi(i) = i$ for every i .

Thus H_w embeds into $\mathbb{Z}$ or into a finite cyclic group. Since H_w is finite, in the former case it is trivial; in the latter it is cyclic. $\square$

Theorem 6.4 (Sufficiency in ZFC). Let λ be uncountable with $\text{cf}(\lambda) = \omega$, and let E be a finite coded S_m -set satisfying (6). There is an equivariant rule

$$L : \text{Inj}(m, Q_\lambda) \longrightarrow E$$

definable from λ and one parameter of rank below $\omega + 10$.

Proof. When $m = 0$, the domain has one element and the condition is exactly $E \neq \emptyset$; choosing the least label proves the assertion. Assume $m \geq 1$. Choose a well-order $\leq$ of the set $\mathcal{A}_m^\omega$ of all incidence words. This is a low-rank set: the alphabet is finite, the words are coded by reals, and a well-order of the reals has rank below $\omega + 10$. Fix also the finite natural order on the labels in E .

Given $\vec{q} \in \text{Inj}(m, Q_\lambda)$, choose representatives $a_i \in q_i$. The union of finitely many cofinal ω -sets has order type ω : it is cofinal, and below any of its points each of the finitely many representatives has only finitely many points. Enumerate this union increasingly as $\langle \beta_n : n < \omega \rangle$ and form the word

$$w_n = \{i < m : \beta_n \in a_i\}.$$

Each column is infinite. Two columns agree eventually exactly when the associated representatives differ finitely; hence our tuple gives pairwise eventually different columns.

Changing finitely many points of the representatives changes only a finite initial part of the merged ordered union and of its letters. Accordingly it leaves the tail class $[w]_{\text{tail}}$ unchanged. Permuting the input tuple acts on this class by the same permutation of letters.

Consider the finite S_m -orbit of $[w]_{\text{tail}}$. Among *all words* representing its tail classes, let w_* be the $\leq$ -least. This choice depends only on the orbit. The stabilizer H_{w_*} is cyclic by Lemma 6.3. If h generates it, a label fixed by h exists by (6); that label is fixed by all of H_{w_*} . Let e_* be the least such label. The trivial-stabilizer case is included, since (6) for the identity says $E \neq \emptyset$.

Choose any $\sigma \in S_m$ with $[w]_{\text{tail}} = \sigma \cdot [w_*]_{\text{tail}}$ and set

$$L(\vec{q}) = \sigma \cdot e_*.$$

This is independent of the choice of σ : if σ_1 and σ_2 give the same class, $\sigma_2^{-1}\sigma_1 \in H_{w_*}$ fixes e_* . It is independent of the chosen representatives by tail invariance. Applying a permutation π changes σ to $\pi\sigma$, so the rule is equivariant.

Although representatives were used to describe the construction, the proof of independence supplies a unique value. The resulting function is definable from the indicated low-rank well-order and the finite action code. $\square$

Corollary 6.5 (Exact classification). In the Prikry extension of Theorem 1.1, the existence of a permitted definable equivariant finite-label rule is equivalent to (6). The same is true with permitted parameters restricted to V_κ and q_0 .

The condition does not demand a label fixed by the whole group. For example, let S_3 act on the disjoint union of its three-point natural action and its two-point sign action. A transposition fixes a point in the first part, a three-cycle fixes both points in the second, and the identity fixes every point. Nevertheless there is no global fixed point. The five-label rule exists. In contrast, the natural S_n -action on $[n]^r$, $0 < r < n$, fails the condition because an n -cycle fixes no such subset.

7 Finite families of ultrafilters

Write $\text{UF}(q)$ for the set of ultrafilters on the *full ambient power set* $\mathcal{P}(q)$. Principal ultrafilters are included unless explicitly excluded. This is different from an internal normal measure on an ordinal in a smaller inner model.

For $a, b \in q$, define the finite-change index

$$\text{ind}_a(b) = |a \setminus b| - |b \setminus a| \in \mathbb{Z}. \quad (7)$$

It satisfies the cocycle identity

$$\text{ind}_a(c) = \text{ind}_a(b) + \text{ind}_b(c) \quad (a, b, c \in q). \quad (8)$$

Indeed, choose an ordinal above all three pairwise symmetric differences. Below this ordinal the identity reduces to cancellation of finite cardinalities, and above it all three sets agree.

For $h \geq 1$, the phase cells

$$P_r^a(h) = \{b \in q : \text{ind}_a(b) \equiv r \pmod{h}\}, \quad r < h, \quad (9)$$

partition q . Every ultrafilter on q contains exactly one of these finitely many cells.

Theorem 7.1 (No finite definable family on the generic class). In $V = W[c]$ of Theorem 1.1, no nonempty finite $\mathcal{F} \subseteq \text{UF}(q_0)$ belongs to $\text{OD}_{W \cup \{q_0\}}$.

Proof. Suppose a condition forces such a uniquely defined family to have size $M \geq 1$, and set $h = M + 1$. In a generic presentation $W[d]$, with $a = \text{ran}(d)$ and $q_0 = [a]$, observe the finite nonempty set

$$R(d) = \{r < h : P_r^a(h) \in D \text{ for some } D \in \mathcal{F}\}.$$

Each ultrafilter selects exactly one phase, so $1 \leq |R(d)| \leq M$.

Pure decision fixes $R(d)$ to a ground-model value R_* below some direct extension (s, A) . Delete the first point after s , obtaining $a^- = a \setminus \{\beta\}$. Equation (7) gives

$$\text{ind}_{a^-}(b) = \text{ind}_a(b) - 1 \quad \text{for every } b \in q_0.$$

Thus $P_r^{a^-}(h) = P_{r+1}^a(h)$, with residues modulo h , and

$$R(d^-) = R(d) - 1 \pmod{h}.$$

The ambient universe, q_0 , and the definable family are unchanged. Both sequences satisfy the deciding condition, so $R_* = R_* - 1$. A nonempty subset invariant under this transitive cycle is all of h . This contradicts $|R_*| \leq M < h$. $\square$

Corollary 7.2 (Finite global kernel families). There is no nonempty finite family, definable from the same parameters, of functions K on Q_κ such that $K(q) \in \text{UF}(q)$ for every q .

Proof. Evaluate every member of the family at the permitted parameter q_0 . Its image is a nonempty finite definable subfamily of $\text{UF}(q_0)$, contrary to Theorem 7.1. $\square$

The theorem covers a family with no definable individual ultrafilter. It also covers principal ultrafilters, so there is no hidden nonprincipality assumption in the obstruction.

8 A canonical countable ultrafilter family

We now work in arbitrary ZFC with an uncountable λ of countable cofinality. No Prikry forcing, rigidity, or elementary embedding is assumed.

For $a \in D_\lambda$, let $a^{(n)}$ be the set obtained by removing its first n points. Fix a nonprincipal ultrafilter u on ω . For $a \in q$, define

$$D_{u,a} = \{X \subseteq q : \{n < \omega : a^{(n)} \in X\} \in u\}. \quad (10)$$

This is the pushforward of u under the injection $n \mapsto a^{(n)}$.

Lemma 8.1 (Tail synchronization). For $a, b \in q$, put $d = \text{ind}_a(b)$. For all sufficiently large n ,

$$b^{(n)} = a^{(n+d)}. \quad (11)$$

Every integer d occurs as $\text{ind}_a(b)$ for some $b \in q$.

Proof. Choose a point beyond $a \triangle b$. Removing k_a points from a and k_b points from b gives a common tail. The difference of these two finite counts is

$$k_a - k_b = |a \setminus b| - |b \setminus a| = d.$$

For $n \geq k_b$, remove a further $n - k_b$ points from that common tail. This yields (11).

For $d \geq 0$, take $b = a^{(d)}$. For $d = -k < 0$, add k distinct ordinals outside a . There are arbitrarily large finite sets of such ordinals because a is countable and λ is uncountable. Adding finitely many points preserves cofinality and order type ω and gives index $-k$. $\square$

Theorem 8.2 (The countable ultrafilter orbit). For every $q \in Q_\lambda$, put

$$\mathcal{U}_u(q) = \{D_{u,a} : a \in q\}. \quad (12)$$

Then:

1. Every member is a nonprincipal ultrafilter on q .
2. $\mathcal{U}_u(q)$ is countably infinite and belongs to $\text{OD}_{u,q}$.
3. The map $T : q \rightarrow q$, $T(b) = b^{(1)}$, induces by pushforward a bijection of $\mathcal{U}_u(q)$. Its integer powers act freely and transitively on that family.

The function $q \mapsto \mathcal{U}_u(q)$ is definable from u and λ .

Proof. Ultrafilter and nonprincipality. The inverse-image operation under $n \mapsto a^{(n)}$ preserves complements and finite intersections. Thus (10) defines an ultrafilter. Since the map is injective, the preimage of a singleton has at most one point; a nonprincipal u contains no finite set. Hence $D_{u,a}$ is nonprincipal.

At most countably many values. Fix one representative $a \in q$ temporarily. If $b, b' \in q$ have the same index d relative to a , then by Lemma 8.1 the two sequences $\langle b^{(n)} : n < \omega \rangle$ and $\langle b'^{(n)} : n < \omega \rangle$ agree eventually. The inverse images of any $X \subseteq q$ therefore differ finitely. A nonprincipal ultrafilter is insensitive to finite changes, so $D_{u,b} = D_{u,b'}$. There is at most one value for each integer d .

Different indices give different values. For a positive integer h , let $r_u(h) < h$ be the unique residue whose class in ω belongs to u . The index identity gives

$$\text{ind}_a(b^{(n)}) = \text{ind}_a(b) + n = d + n.$$

Consequently the unique phase cell of the partition (9) selected by $D_{u,b}$ is

$$P_{d+r_u(h) \bmod h}^a(h). \quad (13)$$

If $d \neq e$, choose $h > |d - e|$. Then these selected phase cells for d and e are different, so the corresponding ultrafilters differ. By the last part of Lemma 8.1, every integer is realized. Thus the family has exactly $\aleph_0$ members.

The integer action. For any $X \subseteq q$,

$$X \in T_*(D_{u,b}) \iff \{n : T(b^{(n)}) \in X\} \in u \iff \{n : b^{(n+1)} \in X\} \in u.$$

The last condition says $X \in D_{u,b^{(1)}}$. Hence $T_*(D_{u,b}) = D_{u,b^{(1)}}$. Relative to the temporary base a , this increments the integer index by one. Since all integer indices occur uniquely at the ultrafilter level, T_* is a bijection and its powers form a free transitive $\mathbb{Z}$ -action. The definition of T_* itself did not use a .

Finally (12) quantifies over all $a \in q$; it makes no choice of a representative. Equations (10) and (12) therefore prove the claimed definability, both locally and as a function of q . $\square$

Corollary 8.3 (Exact local width). In the Prikry extension of Theorem 1.1,

$$\min\{|\mathcal{F}| : \emptyset \neq \mathcal{F} \subseteq \text{UF}(q_0), \mathcal{F} \in \text{OD}_{V_\kappa \cup \{q_0\}}\} = \aleph_0. \quad (14)$$

The same minimum is $\aleph_0$ with $\text{OD}_{W \cup \{q_0\}}$ in place of the displayed definability class. More generally, at any class q where nonempty finite $\text{OD}_{V_\lambda \cup \{q\}}$ families of ultrafilters are excluded, the corresponding minimum is $\aleph_0$.

Proof. The lower bound is Theorem 7.1, or the stated finite-family hypothesis. For the upper bound use Theorem 8.2. An ultrafilter on ω has rank below $\omega + 10$, hence is an allowed low-rank parameter. In a Prikry extension a ground-model nonprincipal ultrafilter on ω is still an ultrafilter, since no new subsets of ω have been added. $\square$

Applied to the classes in [1, §13, theorem on finitely additive kernels], this supplies the announced sharp countable refinement in the ultraexacting setting. Only that application imports the earlier finite-family exclusion; the universal upper bound and its Prikry lower bound were proved here.

Countable does not mean definably enumerated

The family $\mathcal{U}_u(q_0)$ is definable and countable in the ambient universe. Its proof of countability temporarily chooses a representative. There is no enumeration of it definable from the permitted parameters in the Prikry situation: the first element of such an enumeration would itself be a definable ultrafilter, contrary to Theorem 7.1.

Nor have we constructed a countable set $\{K_n : n < \omega\}$ of global ultrafilter-valued kernels. We have constructed one *countable-valued* function $q \mapsto \mathcal{U}_u(q)$. Passing from countable fibers to a single countable family of global choices requires an additional simultaneous choice of phases. This paper makes no such claim.

9 The normal-trace inner model

This section gives the inner-model clause used for the exact consistency calibration. It is compatible with the preceding nonexistence results because it concerns a different ultrafilter: a normal measure on κ *inside a smaller model*, rather than an ultrafilter on q_0 measuring every ambient subset of q_0 .

9.1 Relative cone homogeneity

Lemma 9.1 (Ground-model capture). In $V = W[c]$, every set of ordinals in $\text{OD}_{W \cup \{q_0\}}$ belongs to W . For each $p \in W$,

$$\text{HOD}_{p, q_0}^V \subseteq W. \quad (15)$$

Proof. Any two conditions (s, A) and (t, B) have direct extensions with the same reservoir

$$C = (A \cap B) \setminus (\max(s \cup t) + 1),$$

with the evident interpretation when both stems are empty. Their cones are isomorphic by Lemma 3.6. Corresponding generics give the same universe, the same finite-change class, and the same ground-model parameters. Thus the isomorphism preserves truth of every formula whose parameters are ground-model sets, ordinals, and the named generic class.

No two conditions can force opposite values of such a sentence: refining to these isomorphic cones would transport one decision to the other cone, contradicting the opposite decision. Since deciding conditions are dense, the top condition decides every such sentence.

Let $x \subseteq \delta$ be ordinal definable from p, q_0 and ordinal parameters, where $p \in W$ and δ is an ordinal. The assertion that the chosen formula defines a unique subset of δ is true in the given generic extension, hence is forced by the top condition by the preceding argument. Membership of every $\xi < \delta$ is likewise decided by the top condition. The forcing relation is definable in W , so Separation in W gives exactly x .

To obtain (15), let $x \in \text{HOD}_{p,q_0}$. Relativized HOD has Choice and its canonical definable well-order. Its transitive closure of $\{x\}$ can therefore be coded by a well-founded extensional relation on an ordinal, together with a distinguished index for x , with the code ordinal definable from p, q_0 and ordinals. Convert this relation to a set of ordinals by an ordinal pairing code. That code belongs to W by the first part. Its Mostowski collapse, computed in W , is the same transitive structure as in V . Thus $x \in W$. $\square$

We use here the ordinary relativized-HOD theorem: $\text{HOD}_{p,q}$ is a transitive inner model of ZFC. The parameters need not themselves be hereditary members. One obtains the definable well-order by taking least definition codes; closure under the ZF operations follows by relativizing formulas to this definable class, and the well-order restricted to any of its sets witnesses Choice. These facts apply to fixed *set* parameters, not to an added truth predicate.

9.2 Adding a code for all low-rank parameters

Theorem 9.2 (A normal tail measure with full low-rank agreement). Choose in W a bijection $b : \kappa \rightarrow V_\kappa^W$ and put

$$N = \text{HOD}_{b,q_0}^V.$$

Then $N \models \text{ZFC}$, $N \subseteq W$, and $V_\kappa^N = V_\kappa^V$. Moreover

$$U_N = \{A \in \mathcal{P}(\kappa) \cap N : \exists a \in q_0 (a \subseteq^* A)\} \quad (16)$$

belongs to N and is there a nonprincipal normal κ -complete ultrafilter. In particular N regards κ as measurable.

Proof. Since κ is inaccessible in W , $|V_\kappa^W| = \kappa$, so b exists. Ground-model capture gives $N \subseteq W$. For every $x \in V_\kappa^V = V_\kappa^W$, some ordinal $\alpha < \kappa$ satisfies $x = b(\alpha)$. Thus x is ordinal definable from b, q_0 . Every member of its transitive closure has the same property, since it too lies in V_κ . Hence $V_\kappa^V \subseteq N$, giving the asserted rank agreement.

For every $A \in U$, the conditions whose reservoirs are contained in A form a dense set. The generic sequence is therefore almost contained in A . If a ground-model $A \subseteq \kappa$ is not in U , its complement is in U , so the generic sequence is almost disjoint from A and cannot also be almost contained in it. It follows, for every $A \in \mathcal{P}(\kappa) \cap N \subseteq W$, that

$$\exists a \in q_0 (a \subseteq^* A) \iff A \in U.$$

Consequently $U_N = U \cap N$, with the intersection restricted to subsets of κ .

The definition (16) is ordinal definable from b, q_0 , and all its members belong to N . Transitivity of N shows that all members of its transitive closure belong to N as well. Hence $U_N \in \text{HOD}_{b,q_0}^V = N$.

The ultrafilter property and nonprincipality follow from U . If $\langle A_i : i < \mu \rangle \in N$ with $\mu < \kappa$ and every $A_i \in U_N$, the entire sequence belongs to W . Its intersection is in U by completeness in W , and in N by closure in N . Therefore it belongs to U_N . This proves *internal* κ -completeness.

For normality, let $S \in U_N$ and let $f : S \rightarrow \kappa$ be regressive and belong to N . Both objects belong to W . Normality of U gives an ordinal $\alpha < \kappa$ for which $\{\xi \in S : f(\xi) = \alpha\} \in U$. This fiber belongs to N , since $f, S, \alpha \in N$, and hence lies in U_N . Thus normality holds in N . $\square$

The completeness is deliberately internal. In V , the sequence of measure-one sets $\kappa \setminus (c_n + 1)$ has empty intersection. There is no contradiction because this external sequence need not belong to N .

Corollary 9.3 (Rigidity and no short cofinal information). Under Theorem 9.2, there is no cofinal subset of κ of order type less than κ in OD_{b,q_0} . More generally, every nonempty $\mathcal{F} \in \text{OD}_{b,q_0}$ whose elements are cofinal subsets of κ of order type below κ has ambient cardinality at least κ . In particular the usual rigidity conclusion holds relative to V_κ and q_0 . Also

$$q_0 \notin N, \quad q_0 \cap N = \emptyset.$$

Proof. A set of ordinals in OD_{b,q_0} is hereditary in that definability class and hence belongs to N . Such a cofinal subset of short order type would contradict regularity of κ in N .

Suppose a nonempty family as stated has size $\mu < \kappa$. Let $\tau < \kappa$ be the least order type of one of its members and let a be the union of the members with order type τ . Then a is cofinal, is ordinal definable from b, q_0 , and has ambient size at most $\mu \cdot |\tau| < \kappa$. As a subset of the cardinal κ , it therefore has order type less than κ , contradicting the first assertion. This minimum-order-type step does not assume a common size bound on all members of the original family.

A representative in $q_0 \cap N$ would itself be a short cofinal subset. If $q_0 \in N$, transitivity and its nonemptiness would put a representative in N . Both possibilities are excluded. $\square$

10 An exact consistency calibration

Definition 10.1 (The finite-symmetry normal-trace package). The assertion FSNT says that there are λ, b, q satisfying the following conditions in the ambient universe:

1. λ is an uncountable strong limit cardinal of cofinality ω , $b : \lambda \rightarrow V_\lambda$ is a bijection, and $q \in Q_\lambda$.
2. In $N = \text{HOD}_{b,q}$, the tail filter

$$\{A \in \mathcal{P}(\lambda) \cap N : \exists a \in q (a \subseteq^* A)\}$$

is a nonprincipal normal λ -complete ultrafilter and belongs to N .

3. Every nonempty $\text{OD}_{b,q}$ family of short cofinal subsets of λ has ambient size at least λ .
4. For every $1 \leq r < n < \omega$, there is no nonempty finite $\text{OD}_{b,q}$ subfamily of $\text{Sel}_{n,r}(Q_\lambda)$.
5. For every finite coded S_m -set E , an $\text{OD}_{b,q}$ equivariant map $\text{Inj}(m, Q_\lambda) \rightarrow E$ exists exactly when every element of S_m fixes a label.
6. The least ambient size of a nonempty $\text{OD}_{b,q}$ family of ultrafilters on q is $\aleph_0$.

The finite action codes and formula codes are sets; ordinal definability and relativized HOD have their usual first-order definitions. Thus this is an ordinary assertion over ZFC, not a demand for a satisfaction predicate for the universe. The code b ensures that the permitted parameters encompass V_λ : every $p \in V_\lambda$ is $b(\alpha)$ for an ordinal α .

Theorem 10.2 (Equiconsistency).

$$\text{Con}(\text{ZFC} + \text{a measurable cardinal}) \iff \text{Con}(\text{ZFC} + \text{FSNT}).$$

Proof. **Upper bound.** Start with a measurable κ and a normal measure U in a ground model W . Force with $\mathbb{P}_U$. Proposition 3.5 supplies clause (1), taking $\lambda = \kappa$ and a ground-model bijection $b : \kappa \rightarrow V_\kappa^W$. Theorem 9.2 gives clause (2), and Corollary 9.3 gives clause (3).

Because $b \in W$, the selector exclusion in Theorem 5.2 applies in particular to $\text{OD}_{b,q}$. This is clause (4). Theorem 6.2 gives the necessary direction of clause (5). For the sufficient direction use Theorem 6.4; its well-order parameter belongs to V_κ , so it is definable from b and an ordinal.

For clause (6), Theorem 7.1 excludes finite families. Choose a ground-model nonprincipal ultrafilter u on ω . It remains an ultrafilter, belongs to V_κ , and is definable from b and an ordinal. The family $\mathcal{U}_u(q)$ from Theorem 8.2 is therefore a countably infinite member of $\text{OD}_{b,q}$. All six clauses hold simultaneously in the same Prikry extension.

Lower bound. In any model satisfying FSNT, its definable inner model $N = \text{HOD}_{b,q}$ satisfies ZFC. Clause (2) supplies a normal measure in N on the cardinal λ . Hence N is a model of ZFC with a measurable cardinal. Relativizing each fixed axiom and each fixed theorem to this definable inner model gives the usual interpretation and the corresponding relative-consistency implication.

These two arguments are standard forcing and inner-model interpretations, so they establish the stated consistency equivalence without any assumption of transitive models beyond the convenient language used to describe the constructions. $\square$

What the lower bound says—and what it does not

The normal-measure clause supplies the measurable lower bound. We have not proved that the absence of a definable binary selector, by itself, has measurable consistency strength. Nor have we proved that an abstract rigid class has the additional deletion symmetries used in Sections 5–7. The substantive calibration is that adding the entire finite-selector and finite-label package, as well as the sharp countable ultrafilter width, to the normal-trace configuration *does not increase its consistency strength*.

11 Consequences for the research programme

11.1 A question answered, with its quantifiers separated

The supplied synthesis says that the rigidity and normal-trace results can already occur at measurable strength, while its finite-family selector theorem and finite-label necessity have no Prikry-model counterpart yet. Theorems 5.2 and 6.2 provide that counterpart. They answer positively the explicitly asked question about the Prikry extension of a measurable.

The associated phrase “do they follow from rigidity alone?” is a stronger question, not an equivalent reformulation. A Prikry extension has pure decision and cone surgery in addition to a rigid class. Our proofs use those two extra properties. The implication from abstract rigidity alone is not established here.

11.2 The finite-to-countable local threshold is sharp

The synthesis rules out finite definable families of ultrafilters on certain orbit classes. The universal construction in Section 8 shows that no such theorem can be strengthened to exclude all countable families with the same low-rank parameters. The obstruction ends exactly at finite cardinality for the local ultrafilter problem. This sharpness does not need an additional large-cardinal hypothesis.

There is a concrete reason: shifting a nonprincipal ultrafilter on ω produces an infinite countable orbit, while every nontrivial finite periodic quotient detects the shift. A finite family cannot absorb all cyclic

phase changes in the forcing argument, but the full integer orbit can.

11.3 What would distinguish the stronger embedding structure

The present proofs do not establish successor measurability in parameterized HOD models, simultaneous control over the orbit classes of one ultraexacting witness, or the existence of any rank-into-rank embedding. Those additional assertions in the synthesis lie outside the mechanism isolated here. In particular, the result does not lower the consistency strength of ultraexactingness, whose relation to I_0 is the subject of [5].

An effective next mathematical distinction is to separate hypotheses about a *single generic tail with finite surgery* from hypotheses about *many fixed orbit parameters and their successor-cardinal structure*. The finite-selector obstruction alone is not a discriminator between those two kinds of input.

11.4 Remaining questions

Three precise questions remain after this continuation. First, does rigidity of a class, without a Prikry presentation or an embedding, imply the finite-selector obstruction? Second, can a countably infinite definable family of global ultrafilter-valued kernels exist in the ultraexacting setting, or in a normal Prikry extension? The countable-valued construction here does not settle that. Third, can the deletion method be transferred to a broader Prikry-type forcing where direct finite decision holds but arbitrary stem replacements do not preserve genericity? The exact missing interface in that case is Lemma 3.6, not finite combinatorics.

A Proof audit and reproducibility

A.1 Dependency ledger

Conclusion	Inputs actually used
Finite selector families	Normal-measure diagonalization, pure decision, one-point stem deletion, finite restriction tables, and cycle amplification.
Finite-label necessity	Pure finite decision, stem deletion, and the explicit permutation probes.
Finite-label sufficiency	Choice of a low-rank well-order of incidence words and the cyclic-stabilizer lemma; no forcing.
Finite ultrafilter exclusion	Pure finite decision, deletion, the finite-change index, and finite phase partitions.
Countable ultrafilter family	A nonprincipal ultrafilter on ω , synchronization of finite-change tails, and the uncountability of λ .
Normal-trace inner model	Cone homogeneity relative to the generic class, standard relativized HOD, and the original ground normal measure.
Equiconsistency	The preceding construction for the upper bound; the explicit measurable inner model for the lower bound.

A.2 Points where an incorrect proof could look convincing

No chosen member of a finite definable family. The tables form a set. It is this finite set that is decided and shown invariant. No selector is declared definable merely because it belongs to a definable finite family.

No assumed universe automorphism. Stem surgery identifies two generic presentations of the same extension. The invariance of a definable object follows from identical parameters in that extension, not from a nontrivial automorphism of the membership universe.

No assumption that a pure extension is in a fixed generic. The contradiction is a forcing argument below a condition. A fresh generic through its deciding direct extension is used as needed.

No confusion between the generic sequence and its class. Finite deletion changes the sequence. It leaves its finite-change class unchanged. Only the latter is an allowed new parameter in the negative results.

No external completeness claim. The normal measure U_N measures the subsets of κ belonging to N and is complete for sequences in N . The ambient generic sequence witnesses that external countable completeness would be false.

No conversion of a countable-valued map to a countable set of maps. The construction $q \mapsto \mathcal{U}_u(q)$ has countable fibers. An enumeration of every fiber with coherent indices has not been supplied.

No inference of literature novelty from failed searches. The results are new relative to the supplied synthesis. The cited primary sources establish the classical forcing and large-cardinal background; the finite-observation application here is presented with a proof rather than an unsupported priority claim.

A.3 The finite checks

The archive contains `checks/finite_checks.py` and its actual run output. The script exhaustively computes cyclic orbits of small selector tables, checks cyclicity of stabilizers of small periodic incidence words with distinct columns, verifies the five-label S_3 example, and tests the finite-change index identities on finite perturbation data.

The recorded run checked 193,312 selector tables in 28 parameter cases (including 16 instances of the cycle-amplification bound), 20,523 cyclic stabilizers among 23,590 periodic-word candidates, all six permutations in the five-label example, 32,768 index-cocycle triples, and 2,560 deletion identities, each also checked modulo 1 through 8. All checks passed.

These checks can catch sign errors, mistaken action conventions, and failures of small finite cases. They are *not* a verification of forcing, genericity, HOD, normal measures, or consistency. Those claims rest on the written proofs. No Lean verification is claimed; the supplied Lean files served as a reference for the established finite combinatorial interfaces.

A.4 Source scope

The supplied archive contains two LaTeX reports and the Lean development, but not the twenty-seven individual continuation reports mentioned by the synthesis. Accordingly the comparison here is with the actual supplied synthesis, especially its sections on finite choice, finite-label classification, hidden measures, and remaining questions. The source map bundled with this paper records the exact supplied paths and theorem labels used for that comparison.

References

- [1] *Large cardinals at the inconsistency frontier: a synthesis*. Third-round synthesis supplied in Cardinals4.zip, 18 September 2026. Actual path: docs/Large_Cardinals_Synthesis.tex. Especially §9 (finite selector families), §12.5 (finite-label classification), §13 (rigidity, traces, and ultrafilters), and §15 (remaining questions).
- [2] *Large cardinals at the inconsistency frontier: a unified assessment*. Supplied research plan, 18 September 2026. Actual path: docs/Large_Cardinals_Unified_Report.tex.
- [3] T. Benhamou. *Prikry Forcing and Tree Prikry Forcing of Various Filters*. [arXiv:1801.04424v2](#), 2018. Primary reference for normal-measure Prikry forcing, the Prikry property, cardinal preservation, and genericity criteria. The finite-decision and stem-surgery ingredients used here are proved in Section 3.
- [4] J. P. Aguilera, J. Bagaria, P. Lücke. *Large cardinals, structural reflection, and the HOD Conjecture*. [arXiv:2411.11568v4](#), 2025. Background on exacting and ultraexacting cardinals; not an input to the new forcing obstruction.
- [5] J. P. Aguilera, J. Bagaria, G. Goldberg, P. Lücke. *Large cardinals beyond HOD*. [arXiv:2509.10254](#), 2025. Background on the consistency calibration of exacting and ultraexacting cardinals; not used to obtain the measurable upper bound here.